MISSION PATH & DEPLOYMENT/RECOVERY POINTS (KMZ)

+Add to Map KMZ_UVI2_Around_STX_20250306.kmz

Convert KML file to point and polyline feature class layers

KML To Geodatabase (Conversion Tools):

- Input KML File: KMZ_UVI2_Around_STX_20250306
- Output Point Feature Class: KMLpoints_STX2025_WGS9184
- Output Line Feature Class: KMLpolylines_STX2025_WGS9184

Project KML feature class layers to UTM / VIVD coordinate system for calculations (include vertical coordinate system since original layer has VSC: EGM96_height)

Project (Data Management Tools):

- Input Dataset or Feature Class: KMLpoints_STX2025_WGS9184
- Output Dataset or Feature Class: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\KMLpoints_STX2025
- Output Coordinate System: NAD_1983_UTM_Zone_20N / VCS:VIVD09_height
- Geographic Transformation: Leave as default (WGS_1984_(ITRF00)_To_NAD_1983)

Project (Data Management Tools):

- Input Dataset or Feature Class: KMLpolylines_STX2025_WGS9184
- Output Dataset or Feature Class: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\KMLpolylines_STX2025
- Output Coordinate System: NAD_1983_UTM_Zone_20N / VCS:VIVD09_height
- Geographic Transformation: Leave as default (WGS_1984_(ITRF00)_To_NAD_1983)
- ✅ Preserve Shape

Delete data for intermediate surfacing points

Attribute Table KMLpoints_STX2025**:**

- Select all rows for intermediate surfacings (NOT deployment or recovery point) -> Delete selected rows
- If deployment/recovery points have same value in SymbolID column, change value in 1 row (to allow for different symbology points)

Format mission points and path line

Symbology:

- Delete symbol class row for intermediate surfacings in KMLpoints_STX2025
- Format symbology for deployment point, recovery point, and path line

ACOUSTIC DETECTION POINTS (CSV) ----------

+Add to Map interpolated_HUWH_UVI2_Around_STX_20250306.csv

Create point feature class layer from CSV data

XY Table To Point (Data Management Tools):

- Input Table: interpolated_HUWH_UVI2_Around_STX_20250306.csv
- Output Feature Class: interpolated_HUWH_STX2025_WGS1984
- X Field: Longitude
- Y Field: Latitude
- Coordinate System: GCS_WGS_1984 (since coordinates in decimal degrees)

Convert to UTM coordinate system for calculations

Project (Data Management Tools):

- Input Dataset or Feature Class: interpolated_HUWH_STX2025_WGS1984
- Output Dataset or Feature Class: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\GISCALC_HUWH_STX2025
- Output Coordinate System: NAD_1983_UTM_Zone_20N
- Geographic Transformation: Leave as default (WGS_1984_(ITRF00)_To_NAD_1983)

Format detection points on map

Symbology:

- Format detection points with custom glossy white point symbol

SHORELINES ----------

+Add to Map CARIBBEAN_SHORELINES

Format shorelines layer with solid green custom color scheme

Symbology:

- Format with custom green fill and black outline polygon colors

MAP EXTENT BOUNDARY ----------

Create new polygon feature class layer defining map boundary to use for clipping data/raster layers

In Contents View open mission geodatabase
(C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb)

☀**New -> Feature Class:**

- Name: mapextent_STX2025
- Feature Class Type: Polygon
- Spatial Reference > Current XY: NAD 1983 UTM Zone 20N
- Leave rest as default

Edit top ribbon -> Create:

- Templates: Select mapextent_STX2025
- Drawing method: Select rectangle

Make rectangle length/width sides even and ratio close to ~0.77 to match layout

Edit top ribbon -> Modify:

- Tools: Edit Vertices/Scale/Move

66.176°W - 62.841°W = 3.346 deg wide & 18.892°N - 16.317°N = 2.575 deg long
2.575/3.346 = 0.770 ~ 0.773 proportions of layout ANSI E 34”x44”

BATHYMETRIC DEPTH/SLOPE RASTERS ----------

+ Add to Map GBC15a_bathy **and** GBC15a_slope
+ Add to Map USC10m_bathy **and** USC10m_slope

Crop GEBCO bathy (depth) and slope rasters to mission map extent

Extract by Mask (Spatial Analyst Tools):

- Input Raster: GBC15a_bathy
- Input Raster or Feature Mask Data: mapextent_STX2025
- Output Raster:
  C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\GBC15a_bathy_STX2025
- Extraction area: Inside (inside = keep the raster cells that fall inside your mask polygon)
- Environments > Cell Size: Same as layer GBC15a_bathy
- Environments > Snap Raster: GBC15a_bathy (ensures perfect cell alignment)
- Environments > Compression: NONE

Extract by Mask (Spatial Analyst Tools):

- Input Raster: GBC15a_slope
- Input Raster or Feature Mask Data: mapextent_STX2025

- Output Raster: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\GBC15a_slope_STX2025
- Extraction area: Inside (inside = keep the raster cells that fall inside your mask polygon)
- Environments > Cell Size: Same as layer GBC15a_slope
- Environments > Snap Raster: GBC15a_slope (ensures perfect cell alignment)
- Environments > Compression: NONE

Crop USC bathy (depth) and slope rasters to mission map extent

Extract by Mask (Spatial Analyst Tools):

- Input Raster: USC10m_bathy
- Input Raster or Feature Mask Data: mapextent_STX2025
- Output Raster: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\USC10m_bathy_STX2025
- Extraction area: Inside (inside = keep the raster cells that fall inside your mask polygon)
- Environments > Cell Size: Same as layer USC10m_bathy
- Environments > Snap Raster: USC10m_bathy (ensures perfect cell alignment)
- Environments > Compression: NONE

Extract by Mask (Spatial Analyst Tools):

- Input Raster: USC10m_slope
- Input Raster or Feature Mask Data: mapextent_STX2025
- Output Raster: C:\Users\kkeii\OneDrive\Documents\ArcGIS\Files\THESIS_Files\Missions\mission_UVI2_Around_STX_20250306\mission_UVI2_Around_STX_20250306.gdb\USC10m_slope_STX2025
- Extraction area: Inside (inside = keep the raster cells that fall inside your mask polygon)
- Environments > Cell Size: Same as layer USC10m_slope
- Environments > Snap Raster: USC10m_slope (ensures perfect cell alignment)
- Environments > Compression: NONE

Format bathymetric depth raster and slope raster

Symbology:

- GBC15a_bathy_STX2025 + USC10m_bathy_STX2025 shaded relief with SHADEDRELIEF01_magma color scheme, multidirectional + adjusted shading settings, Z scale factor = 1, pixel size power = 0.5, and pixel size factor = 0.5
- GBC15a_slope_STX2025 + USC10m_slope_STX2025 classify symbology type with SLOPEUSE01_trans_blurple color scheme and manual intervals (11 classes: ≤ 1.72, ≤ 3.43, ≤ 5.71, ≤ 8.53, ≤ 11.3, ≤ 14.04, ≤ 16.7, ≤ 21.8, ≤ 30.96, ≤ 45.0, ≤ 90.0 *(*copy symbology from UNclipped slope raster layer to get intervals without manually typing value for each)*)

MPAs AND SEAFLOOR MORPHOLOGY ----------

+Add to Map WDPA_WDOECM_Caribbean_poly
+Add to Map WDPA_WDOECM_Caribbean_point (if no points in mission area, can remove layer)

+Add to Map GRIDArendal_Seafloor_Morphology_Caribbean

Qualitatively examine if/how nearby MPAs and seafloor morphology may be affecting species distributions